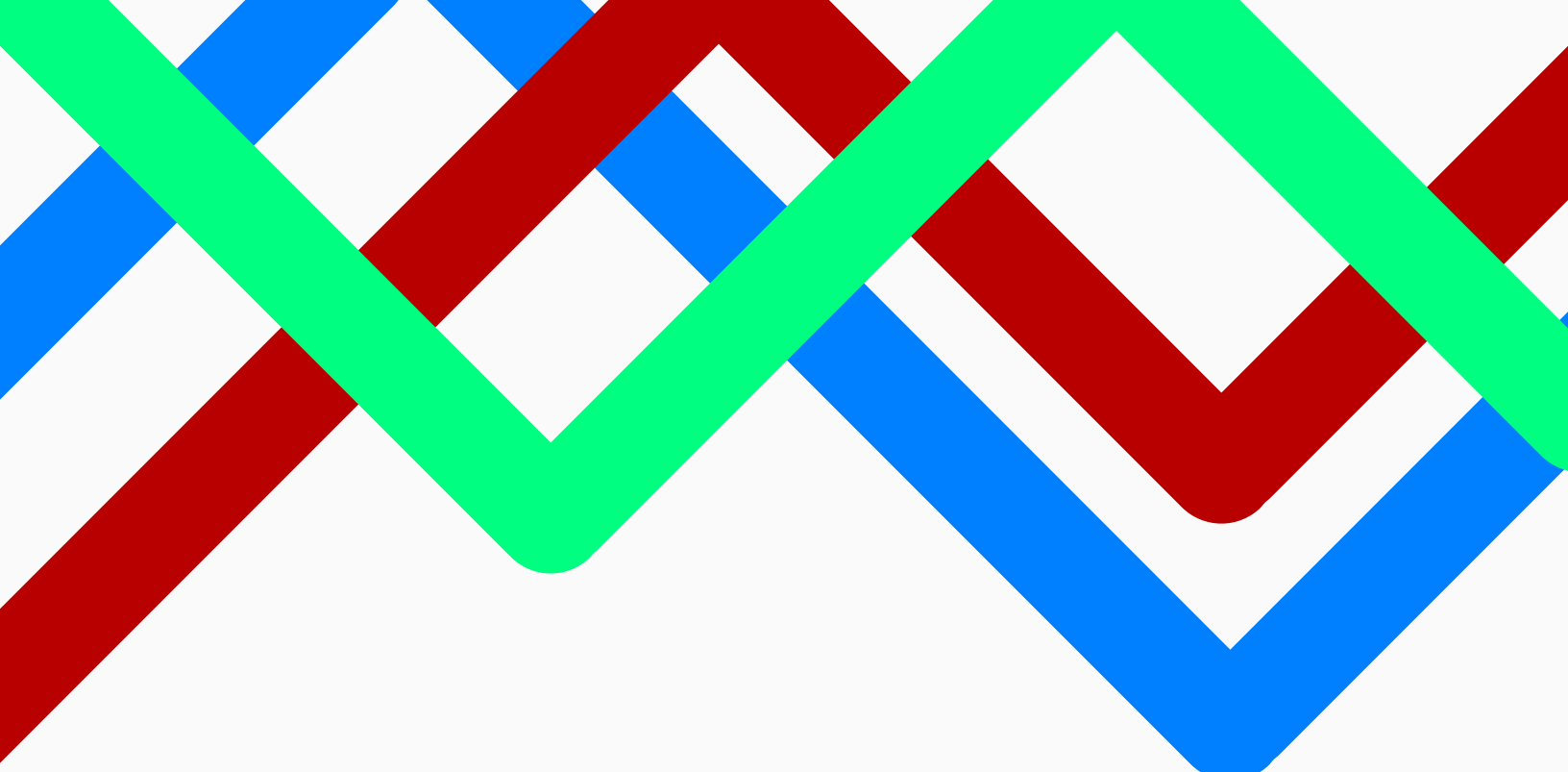




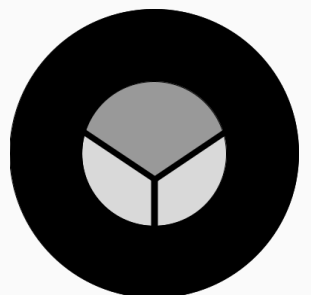
What's the Tea

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“It’s times like these you learn to live again + It’s times like these you give and give again + It’s times like these you learn to love again + It’s times like these time and time again” - Foo Fighters



1 What's the Tea?

Researchers investigated the possible beneficial effect on heart health of drinking black tea and whether adding milk to tea reduces any possible benefit. Twenty-four volunteers were randomly assigned to one of three groups. Every day for a month, participants in group 1 drank two cups of hot black tea without milk, participants in group 2 drank two cups of hot black tea with milk, and participants in group 3 drank two cups of hot water but no tea. At the end of the month, the researchers measured the change in each of the participants' heart health.

- (a) Did the researchers conduct an experiment or an observational study? Explain. Cite at least four significant reasons in your answer.
 - The researchers conducted an experiment, NOT an observational study. This is because of the following four reasons:
 - Randomization: Volunteers are "randomly" selected to one of three groups. This follows the principle of
 - Replication:
 - Control:
 - Presence of variables:
- (b) Why did the researchers include a group who drank hot water but no tea? Explain.
- (c) Is it reasonable to generalize the results of the study beyond the 24 participants? Explain.

The researchers randomly assigned participants to treatment groups.
- (d) What was the reasoning behind that? Explain.
- (e) Describe a method for randomly assigning the 24 volunteers to the three treatment groups. Assume that the researchers wanted an equal number of subjects in each group.
- (f) One aspect of heart health the researchers measured was improvement in blood pressure. The results of the study are graphed below. Is there an association with one or more of the treatment groups and improvement in blood pressure? Explain in 2-3 paragraphs.